

A UNIFIED PHENOMENOLOGICAL DESCRIPTION OF WORK HARDENING AND CREEP BASED ON ONE-PARAMETER MODELS

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Abstract—A phenomenological treatment of plastic deformation is proposed which makes it possible to describe in a unified way the plastic behavior of a material both under dynamic loading and in creep. The treatment is based on the notion of a unique structure parameter that determines the mechanical state of the material. By combining an evolution equation for this structure parameter with a kinetic equation, which relates the strain rate to the stress at a fixed value of the structure parameter, a complete description of plastic deformation is achieved. The evolution is viewed as that towards a steady state defined by a dynamic equilibrium of athermal work hardening (associated with the storage of dislocations) and strain-rate and temperature dependent work softening (associated with the annihilation of dislocations). Analytical solutions of the resulting set of equations are given for two different models based on physically most plausible assumptions concerning the hardening and competing softening processes. The corresponding expressions for the two deformation modes provide a basis for converting stress-strain data into creep data without any adjustable parameters. This is exemplified by a good agreement between a measured creep curve and the one predicted from stress-strain data in the case of room temperature deformation of polycrystalline copper. Commonly used techniques for evaluating creep and work hardening data are critically discussed.

Résumé—Nous proposons un traitement phénoménologique de la déformation plastique qui permet de décrire de manière unifiée la plasticité d'un matériau sous charge dynamique et en fluage. Ce traitement repose sur l'idée d'un paramètre structural unique permettant de déterminer l'état mécanique du matériau. Nous obtenons une description complète de la déformation plastique en combinant une équation d'évolution pour ce paramètre structural avec une équation cinétique reliant la vitesse de déformation à la contrainte, pour une valeur fixée du paramètre structural. L'évolution tend vers un état stationnaire défini par un équilibre dynamique entre un durcissement athermique (associé au stockage de dislocations) et un adoucissement dépendant de la vitesse de déformation et de la température (associé à l'annihilation des dislocations). Nous donnons des solutions analytiques de l'ensemble des équations résultant pour deux modèles différents, qui reposent sur les hypothèses les plus plausibles physiquement et qui concernent les phénomènes du durcissement et de l'adoucissement en compétition. Les expressions correspondantes pour les deux modes de déformation fournissent une base pour convertir les données contrainte-déformation en données de fluage sans ajuster de paramètre. Pour illustrer ceci, nous montrons le bon accord existant entre une courbe expérimentale de fluage et celle qu'on obtient à partir des courbes contrainte-déformation, dans le cas de la déformation de cuivre polycristallin à la température ambiante. Nous discutons de manière critique les techniques utilisées habituellement pour dépouiller les données du fluage et du durcissement.

Zusammenfassung—Es wird eine phänomenologische Behandlung der plastischen Verformung vorgeschlagen, welche es ermöglicht, sowohl das Verfestigungs- als auch das Kriechverhalten eines Materials einheitlich zu beschreiben. Diese Behandlung basiert auf der Vorstellung von einem einzigen Strukturparameter, der den mechanischen Zustand des Materials bestimmt. Eine in sich geschlossene Beschreibung der plastischen Verformung wird möglich bei einer Ergänzung der Evolutionsgleichung für den Strukturparameter durch eine kinetische Gleichung. Letztere legt den Zusammenhang zwischen der Dehngeschwindigkeit und der Spannung bei einem fixierten Wert des Strukturparameters fest. Die Evolutionsgleichung ist von der Art, daß das System einem stationären Zustand zustrebt, der durch ein dynamisches Gleichgewicht der athermischen Verfestigung und der temperatur- und geschwindigkeitsabhängigen Entfestigung bestimmt wird. Analytische Lösungen des entsprechenden Gleichungssystems sind für zwei verschiedene Modelle gegeben, die auf physikalisch plausiblen Annahmen über die konkurrierenden Ver- und Entfestigungsprozesse basieren. Die daraus folgenden Ergebnisse stellen eine Grundlage für die Umwandlung der Verfestigungsdaten in die Kriechdaten ohne Anwendung jeglicher Anpassungsparameter dar. Eine gute Übereinstimmung zwischen einer gemessenen und einer anhand der Verfestigungsdaten berechneten Kriechkurve illustriert die Anwendbarkeit der Methode. Abschließend werden die herkömmlichen Methoden der Auswertung von Verfestigungs- und Kriechdaten diskutiert.

1. INTRODUCTION

In a series of recent papers [1–4], Kocks and Mecking pursued a phenomenological approach to macro-

scopic plasticity of metals that appears very useful in describing plastic flow in different modes of continuous deformation. In particular, the Kocks–Mecking (KM) model is capable of predicting

the creep behavior of a material from its work hardening behavior. The approach is based on the assumption that the kinetics of plastic flow are determined by a *single* structure parameter S representing the *current* structure. The dependence of the flow stress σ on the plastic strain rate $\dot{\epsilon}$ and the absolute temperature T at a given structure of the material is given by the *kinetic equation*

$$\sigma = \sigma(S; \dot{\epsilon}, T). \quad (1)$$

For a complete description of plastic behavior the kinetic equation is complemented with an *evolution equation* which describes the variation of the structure parameter S with strain ϵ at given strain rate and temperature

$$dS/d\epsilon = f(S; \dot{\epsilon}, T). \quad (2)$$

It is a widely used concept of the theory of plasticity and also a natural consequence of the KM model that, as the deformation progresses, the structure parameter S evolves towards a *saturation value* S_s . Accordingly, the flow stress σ as determined by equation (1) will tend to a saturation, or steady state, value σ_s . Since the quantity S_s is a function of $\dot{\epsilon}$ and T , also σ_s can be expressed as $\sigma_s = \sigma_s(\dot{\epsilon}, T)$ so that in the expressions for *steady state* the structure parameter does not appear explicitly. In the philosophy of the KM model the entire stress-strain curve can be viewed as a "transient" of the flow stress from some initial value to the saturation value corresponding to some equilibrium structure at given $\dot{\epsilon}$ and T . In this sense, the stress-strain curve is related to the transient creep range in a creep test. This concept is well grounded on experimental evidence such as reported in Refs [1-4].

The Kocks-Mecking approach is of a restricted applicability for it is generally expected that the kinetic equation may be governed by more than one structure parameters evolving with different rates [5]. Each of them will assume, after a certain transient strain, its stationary value. Certainly, the latter depends on the current values of all "slower" structure parameters (still varying towards *their* saturation). Therefore, as deformation proceeds, more and more structure parameters will reach stationary values so that eventually, at sufficiently large strains, only the slowest "relaxation mode", associated with the structure parameter with the largest transient strain, will survive. In this sense, all the others can be considered as "instantaneously" adapting to the current value of the slowest in the hierarchy of the structure parameters by assuming their stationary values which depend on the current value of the slowest one as well as on $\dot{\epsilon}$ and T .

It is this most slowly varying structure parameter that is chosen as the only governing parameter S in the *one-parameter approach* of Kocks and Mecking. Evidently this approach is only valid for deformation modes whose characteristic strains largely exceed the transient strains for the relaxation of all relevant

structure parameters others than S . This typically applies for continuous deformation (i.e. deformation without abrupt changes of the strain rate or stress), e.g. in dynamic loading with constant strain rate and in creep at constant stress or constant load. The KM model does not account for the short-scale transient behavior such as the occurrence of a yield point following abrupt changes in strain rate in f.c.c. metals as has been discussed in Refs [3] and [5]. Also, such processes as strain rate cycling, cyclic creep, and fatigue have to be excluded from consideration in the framework of the one-parameter treatment of the KM model. This particularly applies to the cases when the strain per cycle becomes comparable with the transient strains of neglected structure parameters, which, for low temperature deformation, are typically of the order of percent.

In the present paper, the one-parameter approach of Kocks and Mecking is given an exact analytic treatment. Furthermore, the original KM model is extended to cover other physically plausible mechanisms of athermal work hardening. The basic equations underlying this treatment are formulated in Section 2. The analytical results which constitute the kernel of the paper are contained in Section 3. Section 4 deals with the methods of evaluating the work hardening and creep data. In Section 5, the recipe for deducing the creep behavior of a material from its work hardening behavior is applied to the case of room temperature deformation of polycrystalline copper.

2. FUNDAMENTAL RELATIONSHIPS

In the KM model, the average dislocation density ρ is considered as the governing parameter S . Equivalently, one can choose as S the mechanical strength of the obstacles to dislocations glide, $\hat{\sigma}$, which is related to the micromechanical parameter ρ via

$$\hat{\sigma} = \alpha G b \rho^{1/2}. \quad (3)$$

Here G is the shear modulus and b the magnitude of the Burgers vector of dislocations; α is a numerical constant of order unity [3]. The quantity $\hat{\sigma}$ is operationally defined as the flow stress at a given dislocation density measured at "reference conditions" viz. at a reference temperature and a reference strain rate. These can be arbitrarily chosen but once chosen remain fixed so that the state of a material at every moment of deformation under test conditions can be characterized, at least in principle, by bringing it to the reference conditions and measuring its flow stress. The kinetic equation, equation (1), that interrelates the stress σ with the strain rate $\dot{\epsilon}$ at a given temperature for a given value of the "hardness parameter" $\hat{\sigma}$ (i.e. for a given state of the specimen) is conventionally represented as

$$\sigma/\hat{\sigma} = (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m}. \quad (1)$$

In this equation one can easily recognize an Arrhenius equation rewritten in the power-law form merely for convenience. This is justified [1, 6] provided that $m \equiv (\partial \ln \dot{\epsilon} / \partial \ln \sigma)_{T, \epsilon} \gg 1$. (For room temperature deformation of f.c.c. metals, m typically ranges between 200 and 500.) The exponent m is generally a function of temperature, as is the normalization parameter $\dot{\epsilon}$. It should be remarked that at $m \rightarrow \infty$, σ is indistinguishable from $\hat{\sigma}$. Although it has been pointed out [3] that m might vary in the course of deformation, this is disregarded here which, however, does not cause any remarkable error in the present context.

The evolution equation for $\hat{\sigma}$ can be derived by specifying the evolution of the dislocation density ρ [1-3]. To this end, we use the concept of hardening to be controlled by the competition of storage and annihilation (rearrangement) of dislocations which processes are assumed to superimpose in an additive manner [1-4]. In the KM model, the equation for the evolution of ρ reads

$$d\rho/d\epsilon = k_1 \rho^{1/2} - k_2 \rho. \quad (4)$$

The production term, $k_1 \rho^{1/2}$, is associated with the athermal storage ($k_1 = \text{const.}$) of moving dislocations which become immobilized after having travelled a distance proportional to the average spacing between the dislocations, $\rho^{-1/2}$; the process believed to account for the athermal hardening (stage II). Accordingly, k_1 can be evaluated [1-4] as $2(\theta_{II}/G)(ab)^{-1}$, where θ_{II} is the slope of the stress-strain curve in stage II, $\theta_{II} \cong G/200$. The second term, $k_2 \rho$, is associated with dynamic recovery (stage III) which is assumed to follow the first order kinetics, i.e. to be linear in ρ . Since dynamic recovery is thermally activated, k_2 is a function of temperature and strain rate, $k_2 = k_2(\dot{\epsilon}, T)$.

By combining equations (3) and (4) one readily obtains the evolution equation for the structure, or hardness, parameter $\hat{\sigma}$

$$d\hat{\sigma}/d\epsilon = \theta_0(1 - \hat{\sigma}/\hat{\sigma}_s) \quad (2a)$$

with

$$\theta_0 = \alpha G b k_1 / 2 \quad (5)$$

and

$$\hat{\sigma}_s = \alpha G b (k_1/k_2). \quad (6)$$

According to equation (2a) the hardness parameter $\hat{\sigma}$ evolves in the course of plastic deformation towards the steady state value $\hat{\sigma}_s$, i.e. towards a state where the hardness parameter does not vary with the strain, $d\hat{\sigma}/d\epsilon = 0$.

It is recognized that θ_0 is independent of the strain rate and depends on temperature through the temperature dependence of G only. The temperature and strain rate dependence of k_2 , on the other hand, gives rise to the corresponding dependence of $\hat{\sigma}$, which is

assumed to have a power-law form suggested by experimental evidence [1-4]

$$\hat{\sigma}_s/\hat{\sigma}_0 = (\dot{\epsilon}/\dot{\epsilon}_0)^{1/n}. \quad (7a)$$

This form is equivalent to assuming $k_2 \sim (\dot{\epsilon}/\dot{\epsilon}_0)^{-1/n}$.

In equation (7a), the normalization constant $\hat{\sigma}_0$ is the saturation value of the hardness parameter associated with a reference strain rate $\dot{\epsilon}_0$ that can be arbitrarily chosen. It should be emphasized that equation (7a) is of empirical nature. Its relationship to the dependence between the strain rate, stress, and temperature for steady state creep (see below) suggests that it actually covers both the low-temperature and the high-temperature ranges. In the low-temperature range, n is temperature dependent following the predictions of the cross-slip theory [7] while $\hat{\sigma}_0$ and $\dot{\epsilon}_0$ are constant. In the high-temperature range n is constant (typically $n = 4$) while $\dot{\epsilon}_0$ bears an Arrhenius-type temperature dependence and is proportional to $\exp(-Q/kT)$ with Q the activation energy for self-diffusion. The latter circumstance is immaterial as long as constant temperature conditions are considered. The above form of the recovery term implies that the static (or time) recovery of the Bailey-Orowan type can be neglected.

The assumption that the mean free path of dislocations is proportional to $\rho^{-1/2}$ has to be abandoned when the spacing between impenetrable obstacles (or sinks for dislocations) which determine the mean free path is a geometrically determined quantity. This can be the case, for instance, in particle-strengthened or very fine-grained materials where the mean free path is prescribed by the particle spacing or by the grain size, respectively. An appropriate assumption for this case is the mean free path to be constant. Consequently, the term describing the athermal storage rate becomes constant so that the adequate evolution equation is

$$d\rho/d\epsilon = k - k_2 \rho \quad (8)$$

where the constant k is determined by the now constant mean free path $d^\dagger$

$$k = (bd)^{-1}.$$

The evolution equation for $\hat{\sigma}$, equation (2a), is then replaced by a modified one

$$d\hat{\sigma}^2/d\epsilon = (k\alpha^2 b^2) G^2 [1 - (\hat{\sigma}/\hat{\sigma}_s^*)^2] \quad (2b)$$

where the steady state value of the hardness parameter corresponding to $d\hat{\sigma}/d\epsilon = 0$ is given by

$$\hat{\sigma}_s^* = \alpha G b (k/k_2)^{1/2}. \quad (9)$$

[Here and henceforth, an asterisk refers to the modified model based on equation (8).] The strain rate dependence of $\hat{\sigma}_s^*$ stemming from that of k_2 can be represented in the form analogous to equation (7a)

$$\hat{\sigma}_s^*/\hat{\sigma}_0^* = (\dot{\epsilon}/\dot{\epsilon}_0)^{1/n^*} \quad (7b)$$

[†]The constancy of the dislocation storage rate in an evolution equation of the form of equation (8) has been also suggested by Bergström [8]. His interpretation of the parameters entering equation (8) is, however, different from ours.

with

$$n^* = 2n, \quad \dot{\sigma}_s^* = \dot{\sigma}_s (\alpha G b k / k_1 \dot{\sigma}_s)^{1/2}.$$

[Equations (6) and (9) have been used in obtaining the latter relationships].

The fact that the exponents in equations (7a) and (7b) are different implies that the temperature and strain rate dependence of $\dot{\sigma}$ is different in the two models due to the different kinetics of athermal dislocation storage. That means that this dependence must be different in a pure (or solution-hardened) material and in a particle-strengthened material, even when the thermally assisted recovery process remains the same.

Equations (1'), (2a) and (7a) underlying the KM model as well as equations (1'), (2b) and (7b) constitute closed sets of equations which provide two competing ways of describing plastic flow in any given mode of deformation (with the above mentioned limitation to continuous deformation). As the original KM model, the modified model can be easily treated analytically. It is one of the aims of the present communication to provide an analytical basis for distinguishing experimentally between these competing models. To this goal, we will compare below the predictions of both models with regard to the shape of the stress-strain curve in dynamic loading as well as to the variation of the strain rate with strain in primary creep. Furthermore, it will be shown how, with the models under consideration, the creep behavior of a material can be deduced from its work hardening behavior. A possible generalization of the two models will be finally suggested.

3. ANALYTICAL DESCRIPTION OF STRESS-STRAIN CURVES AND CREEP CURVES

At constant temperature, equation (1) can be represented in the differential form

$$d\sigma = \left(\frac{\partial \sigma}{\partial \ln \dot{\epsilon}} \right)_s d \ln \dot{\epsilon} + \left(\frac{\partial \sigma}{\partial S} \right)_i dS. \quad (10)$$

By using a particular realization of the general equation (1) in equation (1') and by substituting the corresponding partial derivatives into equation (10) one obtains for the rate of variation of stress with the strain

$$\frac{d\sigma}{d\epsilon} = \frac{\sigma}{m} \cdot \frac{d \ln \dot{\epsilon}}{d\epsilon} + \frac{\sigma}{\dot{\sigma}} \cdot \frac{d\dot{\sigma}}{d\epsilon}. \quad (11)$$

In equation (11) the rate of variation of the hardness parameter, $d\dot{\sigma}/d\epsilon$, in the KM model and in the modified model is given by equations (2a) and (2b), respectively. Equation (11), together with equation (2a) or (2b), provides a simple description of plastic behavior under dynamic loading and in creep.

For dynamic loading ($\dot{\epsilon} = \text{const.}$), equation (11) yields the following expressions for the hardening coefficient θ defined as the slope of the stress-strain curve.

In the KM model

$$\theta = \theta_0 (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m} [1 - (\sigma/\sigma_s)^2], \quad (12a)$$

in the modified model

$$\theta\sigma = \frac{1}{2}(k\alpha^2 b^2) G^2 (\dot{\epsilon}/\dot{\epsilon}_0)^{2/m} [1 - (\sigma/\sigma_s)^2]. \quad (12b)$$

Here the steady state values of the applied stress σ (i.e. those at which θ vanishes) have been introduced for the two models under consideration

$$\sigma_s = \dot{\sigma}_s (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m} \cdot (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m} \cong \dot{\sigma}_s (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m+1/m} \quad (13a)$$

$$\sigma_s^* = \dot{\sigma}_s^* (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m} \cdot (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m^*} \cong \dot{\sigma}_s^* (\dot{\epsilon}/\dot{\epsilon}_0)^{1/m^*+1/m} \quad (13b)$$

[The (temperature-dependent) normalization factor $\dot{\epsilon}$ has been replaced here by $\dot{\epsilon}_0$ for brevity. For $m \gg 1$, no serious error is connected with this simplification.]

Both equations (12a) and (12b) describe a saturation behavior of σ towards its steady state value. The functional dependence of the hardening coefficient on stress is, however, different: in the KM model it is represented by a straight line in the θ vs σ coordinates, as distinct from the modified model where a *hyperbolic* behavior of θ with σ follows in the low-stress range. For the modified model the linearity of the $\theta\sigma$ vs σ^2 plot is predicted. The stress dependence of the hardening coefficient following from the KM model and from the modified model is depicted on Figs 1 and 2, respectively. (Both the θ vs σ and the $\theta\sigma$ vs σ^2 representations are shown.) It

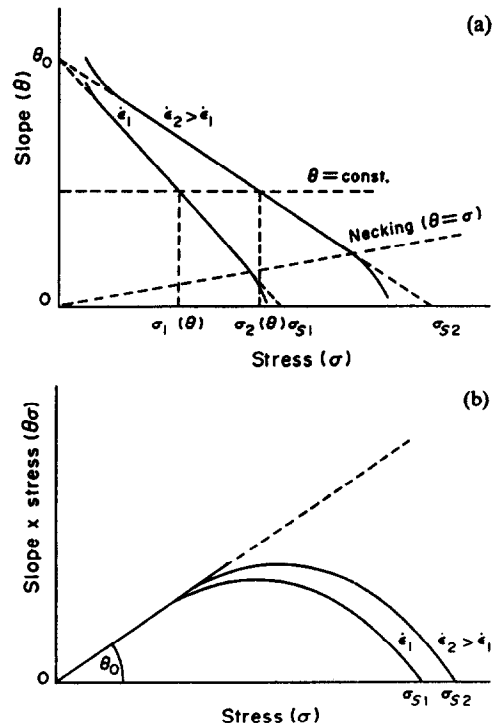


Fig. 1. Stress dependence of the hardening coefficient in the Kocks-Mecking model (schematic): (a) θ vs σ representation; (b) $\theta\sigma$ vs σ representation. On Fig. 1(a), experimentally observed deviations from the linear behavior in the low stress part of the curve as well as on reaching tensile instability (necking) are indicated.

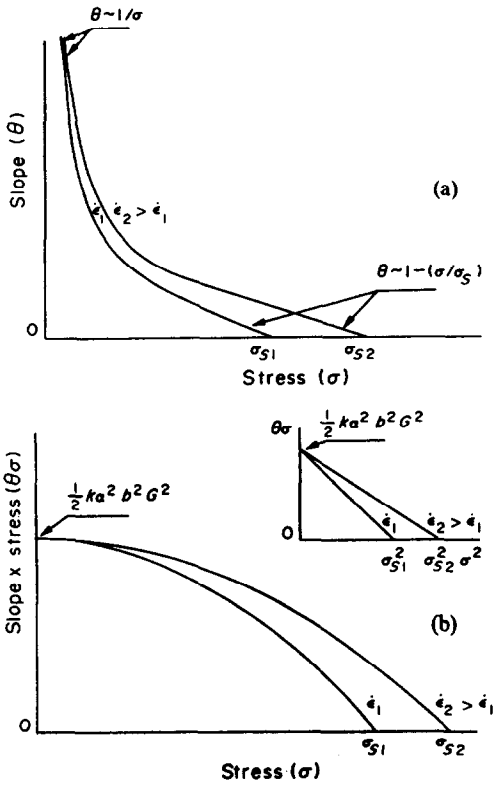


Fig. 2. Stress dependence of the hardening coefficient in the modified one-parameter model (schematic): (a) θ vs σ representation; (b) $\theta\sigma$ vs σ representation; the insert shows this dependence replotted in the straightening coordinates. The asterisks have been omitted.

is recognized that the intercepts of the straight lines on Figs 1(a) and 2(b) with the ordinate axis are respectively taken to be θ_0 and $1/2(k\alpha^2 b^2)G^2$, i.e. their weak strain-rate and temperature dependence [cf. equations (12a) and (12b)] has been neglected, again in view of the assumed largeness of m . It should be noted that the $\theta\sigma$ vs σ plot in the modified model does not pass through the origin, in contrast to the corresponding plot in the KM model. The intercept of the plot with the ordinate axis, proportional to k and thus inversely proportional to the dislocation mean free path d , tends to zero as d increases (e.g. with increasing grain size or particle spacing).

At constant strain rate, equations (12a) and (12b) can be easily integrated to yield

$$\frac{\sigma - \sigma_i}{\sigma_i - \sigma_s} = \exp[-(\varepsilon - \varepsilon_i)/\varepsilon_r] \quad (14a)$$

in the KM model and

$$\frac{\sigma^2 - \sigma_i^{*2}}{\sigma_i^{*2} - \sigma_s^{*2}} = \exp[-(\varepsilon - \varepsilon_i)/\varepsilon_r^*] \quad (14b)$$

in the modified model. Here σ_i and ε_i are some initial stress and strain, respectively (taken, for instance, in the vicinity of the yield point); the characteristic transient strains, or *relaxation strains*, are given by

$$\varepsilon_r = (\sigma_i/\theta_0)(\dot{\varepsilon}/\dot{\varepsilon}_s)^{-1/m} = 2k_2^{-1} \quad (15a)$$

$$\varepsilon_r^* = (k\alpha^2 b^2)^{-1}(\sigma_i^*/G)^2(\dot{\varepsilon}/\dot{\varepsilon}_s^*)^{-2/m} = k_2^{-1}. \quad (15b)$$

It follows from equations (15a) and (15b) that

$$\varepsilon_r^* = \varepsilon_r/2.$$

The relaxation strains, introduced by analogy with relaxation times, determine the rate (in terms of strain, not time!) with which the stress tends to its steady state values. These quantities are obtained from the slopes of the semilogarithmic plots of equations (14a,b). It is interesting to note that the relaxation strains are solely determined by the recovery rate k_2 . An order-of-magnitude estimate of ε_r (and, consequently, of k_2) simply follows from equation (15a)

$$\varepsilon_r \cong \sigma_i/\theta_0.$$

The steady state stress σ_s is a function of strain rate and temperature ranging typically from about $10^{-3} G$ at room temperature to about $10^{-4} G$ at high temperatures while $\theta_0 \cong 10^{-2} G = \text{const.}$ [1-4]. Hence the order of magnitude of the relaxation strain ranges from 10% at room temperature to 1% (or less) at high temperatures. Accordingly, the order of magnitude of k_2 is evaluated as 10 (at room temperature) to 100 (at high temperatures).

A relation of the form of equation (14a) was first suggested heuristically by Voce [9, 10]; equation (14b) is new.

Equations (14a) and (14b) show that the varying stress can approach the level of σ_s (or, respectively, σ_s^*) from below or from above depending on whether σ_i lies below or above the steady state stress. The latter case can be realized e.g. by pre-straining the specimen at a higher strain rate up to a stress level above the steady state stress for the test strain rate. A subsequent deformation with the latter strain rate will then exhibit a "negative" transient towards the steady state.

One of the primary interests in this study is to calculate the strain rate as a function of strain in a creep test at *constant stress*. Equation (11), with $d\sigma/d\varepsilon = 0$ and $d\dot{\varepsilon}/d\varepsilon$ from equation (2a) (KM model) or from equation (2b) (modified model), respectively, reads, after substituting equations (1') and (7)

$$d \ln(\dot{\varepsilon}/\dot{\varepsilon}_s)/d\varepsilon = Q[(\dot{\varepsilon}/\dot{\varepsilon}_s)^{-1/n-1/m} - 1] \quad (16a)$$

(KM model)

$$d \ln(\dot{\varepsilon}/\dot{\varepsilon}_s^*)/d\varepsilon = R[(\dot{\varepsilon}/\dot{\varepsilon}_s^*)^{-2/n^*-2/m} - 1] \quad (16b)$$

(modified model).

Here

$$Q = m\theta_0/\dot{\sigma} \cong m\theta_0/\sigma \quad (17a)$$

$$R = \frac{m}{2}(k\alpha^2 b^2)(G/\dot{\sigma})^2 \cong \frac{m}{2}(k\alpha^2 b^2)(G/\sigma)^2; \quad (17b)$$

the steady state strain rates $\dot{\varepsilon}_s$ and $\dot{\varepsilon}_s^*$ corresponding to $d\dot{\varepsilon}/d\varepsilon = 0$ are given for the two models by

$$\begin{aligned} \dot{\varepsilon}_s &= \dot{\varepsilon}_0^{m/(m+n)} \cdot \dot{\varepsilon}_0^{n/(m+n)} (\sigma/\dot{\sigma}_0)^{nm/(m+n)} \\ &\cong \dot{\varepsilon}_0 (\sigma/\dot{\sigma}_0)^{nm/(m+n)} \end{aligned} \quad (18a)$$

$$\dot{\epsilon}_s^* = \dot{\epsilon}_0^{(m+n^*)} \cdot \dot{\epsilon}_s^{n^*/(m+n^*)} (\sigma/\sigma_{s0}^*)^{mn^*/(m+n^*)} \cong \dot{\epsilon}_0 (\sigma/\sigma_{s0}^*)^{mn^*/(m+n^*)}. \quad (18b)$$

Again the largeness of m as compared to unity has been used to replace $\hat{\sigma}$ by σ in equations (17a) and (17b); to replace $\hat{\epsilon}$ by ϵ_0 in equations (18a) and (18b), the inequality $m \gg n$ has been utilized. Equations (16a) and (16b) are then easily integrated: with the initial condition $\dot{\epsilon} = \dot{\epsilon}_i$ at $\epsilon = \epsilon_i$, the solutions are

$$\frac{\dot{\epsilon}_i^{1/n+1/m} - \dot{\epsilon}_s^{1/n+1/m}}{\dot{\epsilon}_i^{1/n+1/m} - \dot{\epsilon}_s^{1/n+1/m}} = \exp[-(\epsilon - \epsilon_i)/\epsilon_r^{\text{creep}}] \quad (19a)$$

for the KM model and

$$\frac{\dot{\epsilon}_i^{2/n^*+2/m} - \dot{\epsilon}_s^{2/n^*+2/m}}{\dot{\epsilon}_i^{2/n^*+2/m} - \dot{\epsilon}_s^{2/n^*+2/m}} = \exp[-(\epsilon - \epsilon_i)/\epsilon_r^{*\text{creep}}] \quad (19b)$$

for the modified model.

The characteristic relaxation strains for creep, $\epsilon_r^{\text{creep}}$ and $\epsilon_r^{*\text{creep}}$, which determine the rate of variation of $\dot{\epsilon}$ from some initial value $\dot{\epsilon}_i$ towards the steady state value, are expressed as

$$\epsilon_r^{\text{creep}} = \frac{n}{m+n} (\sigma/\theta_0) \text{ (KM model)} \quad (20a)$$

and

$$\epsilon_r^{*\text{creep}} = \frac{n^*}{m+n^*} (k\alpha^2 b^2)^{-1} (\sigma/G)^2 \quad \text{(modified model)} \quad (20b)$$

i.e.

$$\epsilon_r^{*\text{creep}}/\epsilon_r^{\text{creep}} = (\dot{\epsilon}_s^*/\dot{\epsilon}_s)^{1/n}.$$

Equations (19a) and (19b) which describe the primary (transient) creep behavior are quite similar in form with equations (14a) and (14b) for the stress-strain behavior. A comparison of equations (15) and (20) shows that, due to the inequality $n \ll m$, the characteristic relaxation strain in creep turns out to be a small fraction (of order n/m) of the characteristic relaxation strain in dynamic loading. To avoid confusion, we would like to emphasize that the relaxation strains $\epsilon_r^{\text{creep}}$ and $\epsilon_r^{*\text{creep}}$ are not identical with the transient strain for primary creep conventionally defined as the *total* strain acquired by the specimen during the transient towards steady state. The latter quantity should be understood as the sum of the relaxation strain and ϵ_i which implies that the transient strains to reach steady state in dynamic loading and in creep are typically of the same order.

As can be recognized from equations (20a) and (20b) the storage rate constant (k , or k for the KM model or the modified model, respectively) explicitly enters the relaxation strain for creep, as distinct from the transient strain for dynamic loading. This particularly means that by varying the geometrically determined mean free path d (and hence the storage rate $k = (bd)^{-1}$) in particle-strengthened or very fine-grained materials following the evolution equation (8), one can vary the magnitude of the relaxation strain for creep, while the relaxation strain for the

stress-strain curve will remain unchanged. This result can be used for qualitatively checking the applicability of the modified model.

As recognized from equations (20a,b), also in creep the system can approach steady state from above as well as from below. While in conventional primary creep tests the former type of behavior will be observed, the latter can be induced e.g. by abruptly lowering the stress in a sample creeping with a rate close to the steady state strain rate for a given stress. An "inverted" transient curve (approaching steady state from below) will be recorded in this case since, for $m > n$, the system responds to a stress drop by a strain rate lower than the steady state one for the new stress level [5].

As it is well known, transient creep is often observed as *logarithmic creep* represented by the following relations between strain, strain rate, and time t

$$\epsilon - \epsilon_i = A \ln(1 + \beta t)$$

or

$$\dot{\epsilon} = A\beta \exp[-(\epsilon - \epsilon_i)/A] \quad (19c)$$

where A and β are constants. It can be easily seen under what conditions the latter relationship is recovered by equations (19a,b). In the KM model, equation (19a) reduces to

$$\begin{aligned} \dot{\epsilon} &= \dot{\epsilon}_i \exp\left[-(\epsilon - \epsilon_i)/(1/m + 1/n)\epsilon_r^{\text{creep}}\right] \\ &= \dot{\epsilon}_i \exp\left[-(\epsilon - \epsilon_i) \frac{\sigma}{m\theta_0}\right] \end{aligned}$$

having the form of equation (19c) (with $A = \sigma/m\theta_0$, $\beta = m\theta_0\dot{\epsilon}_i/\sigma$), when

$$(\dot{\epsilon}/\dot{\epsilon}_s)^{(m+n)/mn} \gg 1.$$

That means that logarithmic creep corresponds to the initial part of the transient creep curve, far from steady state, when the recovery processes, giving rise to the first term on the right hand side of equation (16a), can be neglected. In this sense, logarithmic creep is related to the stage II part of the stress-strain curve.

Analogously, the condition of negligibility of the recovery term in equation (16b) for the modified model

$$(\dot{\epsilon}/\dot{\epsilon}_s)^{2(m+n^*)/mn^*} \gg 1$$

reduces equation (19b) to the logarithmic creep equation

$$\begin{aligned} \dot{\epsilon} &= \dot{\epsilon}_i \exp\left[-(\epsilon - \epsilon_i) \frac{m+n^*}{2mn^*} \epsilon_r^{*\text{creep}}\right] \\ &= \dot{\epsilon}_i \exp\left[-(\epsilon - \epsilon_i) \frac{1}{2(mk\alpha^2 b^2)^{-1} (\sigma/G)^2}\right]. \end{aligned}$$

A look at equations (19) and (20) which describe primary creep shows that the shape of the $\dot{\epsilon}(\epsilon)$ curve in the transient from an initial value $\dot{\epsilon}_i$ of the strain rate to the steady state strain rate is uniquely given

by three parameters: n, m, k_1 in the KM model and n, m, k in the modified model.

All these parameters can be determined from an analysis of stress-strain curves. The creep behavior of a material can be thus predicted from its strain hardening behavior without any adjustable parameter. In Section 5 it will be done for the case of polycrystalline Cu. Which of the two competing models is to be taken for such data conversion can be decided by checking the fulfilment of their qualitative predictions. In particular, such features of these models as the linearity of the θ vs σ plot in the KM model and of the $\theta\sigma$ vs σ^2 plot in the modified model are most indicative of their validity.

Equations (14a), (14b) and (19a), (19b) give, together with the corresponding expressions for the relaxation strains, a unified description of the stress-strain curve and of the primary creep curve for the two competing models. These equations constitute the major result of this paper. They provide a basis for converting stress-strain data into creep data (and vice versa). The means of determining the parameters involved in the description of the plastic behavior in both deformation modes and thus required for such a conversion are discussed in the next section.

Concluding this section, we would like to stress once again the formal similarity of the derived equations for the stress-strain curve and the primary creep curve. Moreover, equations (13) and (18) which relate the strain rate and the stress in steady state under dynamic loading and in creep, respectively, are identical, as has been repeatedly proven experimentally (cf. e.g. Ref. [11]). This is an immediate consequence of the assumption, underlying the one-parameter approach used, that the hardening behavior in both deformation modes is governed by the same kinetics and evolution laws. These lead to one master differential-equation which can be solved according to the external conditions $\dot{\epsilon} = \text{const.}$ or $\sigma = \text{const.}$, respectively.

4. EVALUATION OF STRAIN HARDENING AND CREEP DATA

4.1. Dynamic loading

The way of determining the exponent m is readily suggested by equation (1'): the magnitude of m can be found from the stress jump upon an "instantaneous" change in strain rate which is supposed to leave the structure and thus δ unchanged

$$1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_{\delta, T} \cong \frac{\ln(\sigma_2/\sigma_1)}{\ln(\dot{\epsilon}_2/\dot{\epsilon}_1)} \quad (21)$$

where σ_1 and σ_2 denote the stresses at the strain rates $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$, respectively, before and after a jumpwise change of the strain rate from $\dot{\epsilon}_1$ to $\dot{\epsilon}_2$ at a fixed temperature T .

Figure 3(a) illustrates the procedure of determining $1/m$. It emphasizes that the stress σ_2 has to be

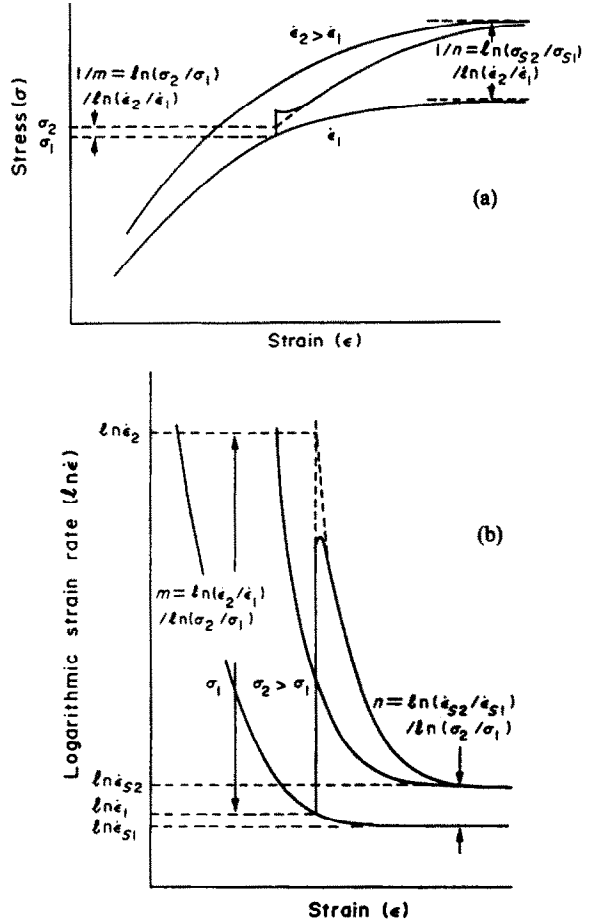


Fig. 3. Determining the exponents m and n : (a) by changing the strain rate in dynamic loading; (b) by changing the stress in creep.

obtained by back-extrapolating the stress-strain curve for $\dot{\epsilon}_2$ that is established after a virtual transient died off. Such transients are associated with the relaxation of neglected structure parameters as discussed in the Introduction.

The sum $1/n + 1/m$ can be determined from the strain rate dependence of the steady state stress [cf. equation (13)] in the KM model

$$1/n + 1/m = (\partial \ln \sigma_s / \partial \ln \dot{\epsilon})_T \cong \frac{\ln(\sigma_{s2}/\sigma_{s1})}{\ln(\dot{\epsilon}_2/\dot{\epsilon}_1)} \quad (22)$$

where σ_{s1} and σ_{s2} are the steady state values of the stress at the strain rates $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$ at a fixed temperature T .

These procedures are common to the two competing models, only that n, σ_s, σ_{s1} and σ_{s2} should be replaced in equation (22) by $n^*, \sigma_s^*, \sigma_{s1}^*$ and σ_{s2}^* for the modified model. The use of the steady state values of the applied stress in obtaining n (or n^*) is connected with an extrapolation since a tensile instability (necking) sets in before steady state is reached [4]. The steady state stress is then found by extrapolating the linear portions of the θ vs σ or $\theta\sigma$ vs σ^2 plots for the KM model and the modified one, respectively, to zero θ . A different procedure of finding $1/n + 1/m$ (or

$1/n^* + 1/m$) can be utilized once the model underlying the strain hardening behavior is identified through a qualitative check of the character of the θ vs σ dependence.

For the KM model, this procedure is suggested by equations (12a) and (13a) from which it follows that this quantity can be determined by taking a section through the data that corresponds to a constant θ [12]. Then

$$1/n + 1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_{\theta, T} \cong \frac{\ln \{ \sigma_2(\theta) / \sigma_1(\theta) \}}{\ln(\dot{\epsilon}_2 / \dot{\epsilon}_1)} \quad (23)$$

Here $\sigma_1(\theta)$ and $\sigma_2(\theta)$ are the stresses corresponding to a given value θ of the hardening coefficients at the strain rates $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$, respectively. The specific value of constant θ chosen is immaterial since

$$\sigma_2(\theta) / \sigma_1(\theta) = \sigma_{\dot{\epsilon}_2} / \sigma_{\dot{\epsilon}_1}$$

at an arbitrary θ , as seen, e.g. from the similarity of the respective triangles on Fig. 1(a). Equations (22) and (23) are thus equivalent.

Analogously, for the modified model, a constant $\theta\sigma$ section through the $\theta\sigma$ vs σ^2 plots for different strain rates can be used to find the exponent n . From equation (12b), such a section corresponds to the fulfillment of condition

$$(\partial \ln \sigma / \partial \ln \dot{\epsilon})_T = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_{\theta\sigma, T}$$

(a weakly strain-rate dependent factor $(\dot{\epsilon}/\dot{\epsilon})^{2/m}$ in equation (12b) being omitted). Accordingly, n^* can be determined from the relationship

$$1/n^* + 1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_{\theta\sigma, T} \cong \frac{\ln \{ \sigma_2(\theta\sigma) / \sigma_1(\theta\sigma) \}}{\ln(\dot{\epsilon}_2 / \dot{\epsilon}_1)} \quad (24)$$

where $\sigma_1(\theta\sigma)$ and $\sigma_2(\theta\sigma)$ are the stresses corresponding to a fixed (arbitrary) value of $\theta\sigma$ at the strain rates $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$, respectively.

Equation (22) [as well as its equivalents, equations (23) and (24) for the KM model and the modified model, respectively] determine, in combination with equation (21), both m and n . As already mentioned, extensive experimental evidence [11] indicates that commonly $m \gg n$ so that effectively m does not enter the expression for the steady state stress and, consequently, that for the hardening coefficient. As a result, one can conclude that the parameter n is relevant in describing the hardening behavior of a material while m is solely involved in describing its response to instantaneous variation of deformation conditions.

Due to the above mentioned occurrence of the tensile instability (necking) preventing a specimen from reaching steady state, the θ vs σ plots for the KM model and $\theta\sigma$ vs σ^2 plots for the modified model should exhibit a systematic deviation from linearity in the lower portions of the curves, as schematically

shown in Fig. 1(a). Such a deviation was indeed observed [13] in θ - σ plots for a number of f.c.c. metals linear in their midpart. Due to the occurrence of necking, the ultimate tensile stress, σ_{UTS} , is often used in place of the steady state stress.

As shown in the Appendix, the relative error $\Delta n/n$ or $\Delta n^*/n^*$ involved in determining the parameter n or n^* by using in equation (22) the UTS instead of the steady state stress is

$$\Delta n^*/n^* = \Delta n/n = (\sigma_s/\theta_0)(\dot{\epsilon}/\dot{\epsilon})^{-1/m} = 2k_2^{-1}. \quad (25)$$

This error ranges from about 10% at room temperature to about 1% at elevated temperatures [cf. the discussion following equation (15a)]. This assessment shows that an *a priori* replacement of the steady state stress by the UTS is generally not legitimate. A simple correction of the data thus obtained is, however, possible with the aid of equation (25).

The above considerations indicate the feasibility of experimentally determining the exponents m and n . Finding out the remaining parameters θ_0 or k in the two models under investigation is straightforward as well. They are obtained by extrapolating to zero stress the linear plots of θ vs σ (KM model) or $\theta\sigma$ vs σ^2 plots (modified model) defined respectively by equations (12a) and (12b). The intercept with the ordinate axis gives respectively

$$\theta_0(\dot{\epsilon}/\dot{\epsilon})^{1/m} \cong \theta_0$$

and

$$(k\alpha^2 b^2/2)(\dot{\epsilon}/\dot{\epsilon})^{2/m} G^2 \cong (k\alpha^2 b^2/2) G^2.$$

4.2. Creep at constant stress

The creep data are as suited for determining the parameters m and n as the stress-strain data. Equation (1') gives a straightforward means of measuring m from the response of the material to an instantaneous change of the applied stress from a level σ_1 to a level σ_2 (cf. Fig. 3(b)). Equation (21) can be immediately used also for this case, where now the quantities $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$ denote the strain rates before and after the stress jump. As illustrated by Fig. 3(b), the latter is understood as the strain rate obtained by back-extrapolating that part of the creep curve for σ_2 where virtual transients have died off. In analogy with equation (22), one further has for the sum $1/n + 1/m$ in the KM model

$$1/n + 1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_T \cong \frac{\ln(\sigma_2/\sigma_1)}{\ln(\dot{\epsilon}_2/\dot{\epsilon}_1)} \quad (26)$$

where $\dot{\epsilon}_{s1}$ and $\dot{\epsilon}_{s2}$ are the steady state values of the strain rate corresponding to the (constant) stresses σ_1 and σ_2 at a given temperature T . The same equation (26) can be used in the modified model, only that the asterisked quantities are to be understood under n , $\dot{\epsilon}_s$, $\dot{\epsilon}_{s1}$ and $\dot{\epsilon}_{s2}$.

One can also produce equations for determining $1/n + 1/m$ or $1/n^* + 1/m$ from the creep data without extrapolating to steady state which are analogous to

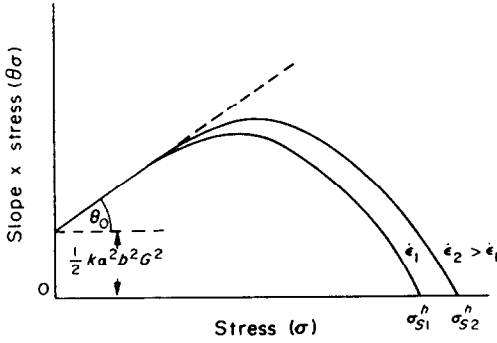


Fig. 4. $\theta\sigma$ vs σ plot for the hybrid model (schematic).

equations (23) and (24) for dynamic loading. To this end, the circumstance can be used that the strain rate $\dot{\epsilon}$ is proportional to its steady state value [within the same accuracy as equations (19a,b) are valid] when a certain combination of $(\partial \ln \dot{\epsilon} / \partial \epsilon)_\sigma$ and σ is kept constant. As seen from equations (16a,b), this combination is

$$\psi = \sigma (\partial \ln \dot{\epsilon} / \partial \epsilon)_\sigma \quad \text{for the KM model} \quad (27a)$$

$$\psi^* = \sigma^2 (\partial \ln \dot{\epsilon} / \partial \epsilon)_\sigma \quad \text{for the modified model.} \quad (27b)$$

This suggests a procedure of deriving $1/n + 1/m$ or $1/n^* + 1/m$ by replotting $\dot{\epsilon}$ vs ϵ curves for given stresses as ψ (or ψ^*) vs ϵ curves with a subsequent constant ψ cut (or constant ψ^* cut) through the family of these plots for various stresses. This yields in the KM model

$$\begin{aligned} 1/n + 1/m &= (\partial \ln \sigma / \partial \ln \dot{\epsilon})_T = (\partial \ln \sigma / \partial \ln \dot{\epsilon})_{\psi, T} \\ &= \frac{\ln(\sigma_2/\sigma_1)}{\ln\{\dot{\epsilon}_2(\psi)/\dot{\epsilon}_1(\psi)\}} \quad (28) \end{aligned}$$

where $\dot{\epsilon}_1(\psi)$ and $\dot{\epsilon}_2(\psi)$ are the strain rates corresponding to a fixed (arbitrarily chosen) ψ at the stresses σ_1 and σ_2 , respectively, for a constant temperature T . This formula can be also used for the modified model by replacing n , σ , and ψ by the corresponding asterisked quantities.

The outlined procedure is designated to avoid extrapolating the creep data to steady state which may be connected with complications due to a tensile instability occurring before the system has a chance of approaching steady state. On reaching the instability condition, the strain rate ceases to decrease and starts increasing. Often in the literature, it is tacitly assumed that the minimum creep rate defined by the instability condition is indistinguishable from the steady state strain rate. The error associated with this (generally unjustified) substitution is evaluated in the Appendix. From this evaluation it follows that the relative error connected with using the minimum creep rate instead of the steady state strain rate in equation (26) is given by

$$\Delta n/n \cong (n/m)(\sigma/\theta_0)(\text{KM model}) \quad (29a)$$

$$\begin{aligned} \Delta n^*/n^* &\cong 2(n^*/m)(\kappa\alpha^2 b^2)^{-1}(\sigma/G)^2 \\ &\quad (\text{modified model}). \quad (29b) \end{aligned}$$

This error is to be compared with that involved in determining n or n^* from the UTS. A comparison of equations (29a,b) with equation (25) shows that the relative error due to the use of the minimum creep rate is smaller than that due to the use of the UTS in dynamic loading tests by a factor of the order of n/m . This indicates that the former error can hardly exceed the order of 1% at room temperature and is even smaller at higher temperatures. This assessment justifies using the minimum creep rate to determine the exponent in the power law of equations (18a,b).

4.3. Creep at constant load

A positive word should be said in this context about the tensile creep test at *constant load* P which is discredited by frequently inappropriate interpretation. Apart from the tensile instability, the corresponding $\dot{\epsilon}(\epsilon)$ -curve would exhibit a minimum followed by an increase due to a continuous increase of the true stress as a result of continuous reduction of the cross-section. It should be emphasized that the condition for this minimum exactly coincides with the necking criterion for creep at constant load [14], $(\partial \ln \dot{\epsilon} / \partial \epsilon)_P = 0$. Thus, one can be certain in this case that the information obtained from the data on the minimum location is really relevant to the intrinsic properties of the material and is not related to the development of deformation instabilities which widely depend on external conditions. It is therefore desirable to produce a formula relating the minimum location to the parameter n (or n^*).

The differential equations describing the variation of $\dot{\epsilon}$ with ϵ at constant load in the two models under consideration can be derived in the same manner as equations (16a,b) for creep at constant stress. Now, the equation $(\partial \ln \sigma / \partial \epsilon)_P = 1$ describing the increase of stress at constant load P in tension should be incorporated in equation (11). The resulting differential equation can be easily integrated in the two models under consideration. The result of this integration can be fairly well represented by the following formulae [15] reminiscent of equations (19a,b):

for the KM model

$$\begin{aligned} \frac{(\dot{\epsilon}/\dot{\epsilon}_s)^{1/n+1/m} - (1 + \sigma_0/\theta_0 + \epsilon)}{(\dot{\epsilon}_i/\dot{\epsilon}_s)^{1/n+1/m} - (1 + \sigma_0/\theta_0 + \epsilon_i)} \\ = \exp \left[- \frac{\epsilon - \epsilon_i}{\epsilon_r^{\text{creep}}} (1 - \sigma_0/\theta_0 - \epsilon_i) \right], \quad (30a) \end{aligned}$$

for the modified model

$$\begin{aligned} \frac{(\dot{\epsilon}/\dot{\epsilon}_s)^{2/n^*+2m} - [1 + 2(\kappa\alpha^2 b^2)^{-1}(\sigma_0/G)^2 + 2\epsilon]}{(\dot{\epsilon}_i/\dot{\epsilon}_s)^{2/n^*+2m} - [1 + 2(\kappa\alpha^2 b^2)^{-1}(\sigma_0/G)^2 + 2\epsilon_i]} \\ = \exp \left\{ - \frac{\epsilon - \epsilon_i}{\epsilon_r^{\text{creep}}} \right. \\ \left. \times [1 - 2(\kappa\alpha^2 b^2)^{-1}(\sigma_0/G)^2 - 2\epsilon_i] \right\}. \quad (30b) \end{aligned}$$

Here $\dot{\epsilon}_s$ and $\dot{\epsilon}_s^*$ denote the "would-be" steady state strain rates, corresponding to the *nominal* (initial) stress σ_0 , for the KM model and the modified model, respectively. It is easily recognized that far from this would-be steady state, i.e. at respectively $(\dot{\epsilon}/\dot{\epsilon}_s)^{1/n+1/m} \gg 1$ and $(\dot{\epsilon}/\dot{\epsilon}_s^*)^{2/n^*+2/m} \gg 1$, both terms added to unity on the left-hand side of equations (30a) and (30b), which stem from the non-constancy of stress, can be neglected. In this range the $\dot{\epsilon}(\epsilon)$ -curve in the constant load test does not differ from that in the constant stress test (and assumes the logarithmic creep form, cf. Section 3). However, the situation changes as $(\dot{\epsilon}/\dot{\epsilon}_s)^{1/n+1/m}$ or respectively $(\dot{\epsilon}/\dot{\epsilon}_s^*)^{2/n^*+2/m}$ approach unity. Now these terms [e.g. σ_0/θ_0 and ϵ in equation (30a)] become important. This results in an increasing branch of the $\dot{\epsilon}(\epsilon)$ -dependence. Indeed, as the right-hand side of equations (30a,b) tends to zero with increasing strain, the strain rate grows due to the term ϵ or 2ϵ on the left-hand side.

A procedure for determining the exponents n or n^* from constant load data is suggested by the following formulae which we give without proof, with reference to a more detailed investigation of constant load creep [15]:

for the KM model

$$1/n + 1/m = (\partial \ln \sigma_0 / \partial \ln \dot{\epsilon}_{\min})_{T,P} \cdot [1 + \sigma_0/\theta_0 + (\partial \epsilon_{\min} / \partial \ln \sigma_0)_{T,P}], \quad (31a)$$

for the modified model

$$1/n^* + 1/m = (\partial \ln \sigma_0 / \partial \ln \dot{\epsilon}_{\min})_{T,P} \cdot \times [1 + 2(k\alpha^2 b^2)^{-1}(\sigma_0/G)^2 + (\partial \epsilon_{\min} / \partial \ln \sigma_0)_{T,P}]. \quad (31b)$$

Here $\dot{\epsilon}_{\min}$ and $\epsilon_{\min}$ are the minimum creep rate in constant load test and the corresponding strain, respectively. These formulae indicate that by varying the nominal stress σ_0 and by measuring the corresponding variation of the minimum location (i.e. of $\dot{\epsilon}_{\min}$ and $\epsilon_{\min}$), the combination $1/n + 1/m (\cong 1/n)$ or $1/n^* + 1/m (\cong 1/n^*)$ can be precisely determined. Evidently this procedure presumes that the parameter characterizing the athermal strain hardening (θ_0 or k for the KM model or the modified model respectively) is known from strain hardening data. Even if these quantities are not exactly known, equations (31a,b) can be used to estimate the error made by a simplistic determination of n or n^* as $(\partial \ln \dot{\epsilon}_{\min} / \partial \ln \sigma_0)_{T,P}$.

5. AN EXPERIMENTAL EXAMPLE

Now that we have discussed the techniques for determining the parameters figuring in equations (18) and (19) which describe the creep behaviour, we can concern ourselves with a concrete experimental example illustrating the main steps in calculating a creep curve from stress-strain data. We consider room temperature deformation of polycrystalline Cu

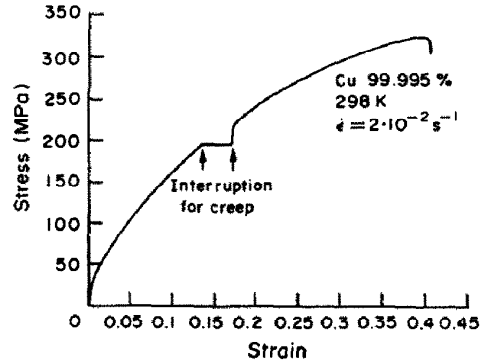


Fig. 5. Stress-strain curve for polycrystalline Cu at room temperature. The horizontal line corresponds to an interruption for a creep test.

(purity 99.995%, average grain size 20 μm). A typical stress-strain curve is presented on Fig. 5. The horizontal part of the curve corresponds to deformation under constant stress creep conditions ($\sigma = 200$ MPa). The creep curve recorded in this test will be compared with that predicted from the stress-strain behavior.

As already indicated in Section 4, the determination of the parameter n (or n^*) involves the stress-strain curves for at least two strain rates. The curve of Fig. 6 corresponding to the strain rate of $2 \cdot 10^{-4} \text{ s}^{-1}$ will be used to complement the curve of Fig. 5 that corresponds to the strain rate of $2 \cdot 10^{-2} \text{ s}^{-1}$. On Fig. 6 also indicated is a strain rate change from $\dot{\epsilon}_1 = 2 \cdot 10^{-4} \text{ s}^{-1}$ to $\dot{\epsilon}_2 = 2 \cdot 10^{-2} \text{ s}^{-1}$ performed in order to determine the stress exponent m by using equation (21). The value of m found in this way is about 230.

While the method of determining m is independent of whether the KM model or the modified model apply, the way in which the stress exponent n (or n^*) is to be found relies basically on the specific model. To check which of the two competing models applies (if at all), we consider the θ vs σ plots associated with the stress-strain curve of Fig. 5 (without the creep portion and a small transient part following it) as well as with the part of the stress-strain curve of Fig. 6 which corresponds to the strain rate of $2 \cdot 10^{-4} \text{ s}^{-1}$. The linearity of these plots which are presented on Fig. 7 substantiates the applicability of the KM

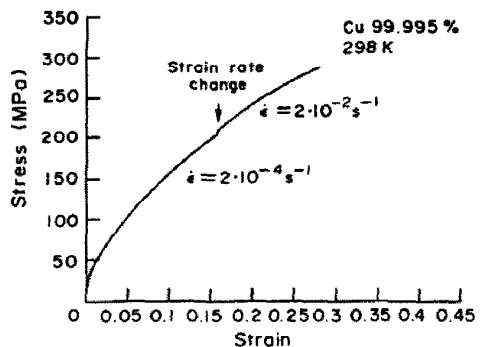


Fig. 6. Strain-rate change for determining m .

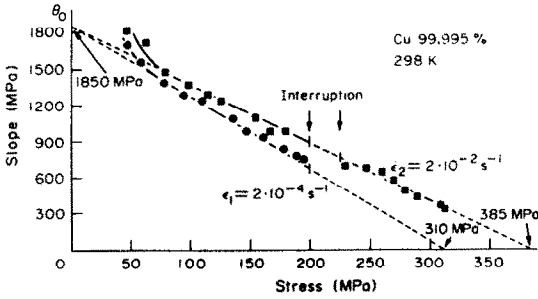


Fig. 7. θ vs σ plots for $\dot{\epsilon}_1 = 2 \cdot 10^{-4} \text{ s}^{-1}$ and $\dot{\epsilon}_2 = 2 \cdot 10^{-2} \text{ s}^{-1}$.

model for the case under consideration. The extrapolation of the plots to zero stress yields the value of $\theta_0 = 1850 \text{ MPa}$, while the extrapolation to zero θ gives the values of the steady state stresses $\sigma_{s1} = 310 \text{ MPa}$ and $\sigma_{s2} = 385 \text{ MPa}$ corresponding to the strain rates $\dot{\epsilon}_1 = 2 \cdot 10^{-4} \text{ s}^{-1}$ and $\dot{\epsilon}_2 = 2 \cdot 10^{-2} \text{ s}^{-1}$, respectively. Now, the stress exponent n can be found with the aid of equation (22): $n = 21.25$.

The validity of the KM model implies that the equation to be used in describing the creep curve is equation (19a), with the steady state strain rate given by equation (18a) and the "relaxation strain" $\epsilon_r^{\text{creep}}$ given by equation (20a). It is convenient to represent the steady state creep rate $\dot{\epsilon}_s$ for a given stress σ in terms of the steady state stress σ_s for an arbitrary strain rate $\dot{\epsilon}$ in a constant strain rate test. Indeed, a comparison of equations (13) and (18) gives the relation

$$\dot{\epsilon}_s = \dot{\epsilon} (\sigma / \sigma_s)^{nm/(m+n)} \cong \dot{\epsilon} (\sigma / \sigma_s)^n. \quad (18c)$$

In particular, we may take $\dot{\epsilon}_2$ and σ_{s2} for $\dot{\epsilon}$ and σ , respectively. In the experiment under consideration, we may, furthermore, set $\dot{\epsilon}_1 = \dot{\epsilon}_2$. Thus, all parameters entering equation (19a) for the creep curve are determined and no adjustable parameter is left.

It is convenient to express the equation for the creep curve in the form

$$\dot{\epsilon}(\epsilon) = \dot{\epsilon}_2 (\sigma / \sigma_{s2})^n \left\{ 1 + (\sigma_{s2} / \sigma - 1) \exp \left(- \frac{\epsilon - \epsilon_i}{\epsilon_r^{\text{creep}}} \right) \right\}^n$$

where ϵ_i is the strain at which the creep test has been started and

$$\epsilon_r^{\text{creep}} = (\sigma / \theta_0)(n/m) \cong 0.01.$$

The creep curve calculated according to this formula is shown, together with the measured curve, in Fig. 8. The agreement between the two curves can be regarded as good.

In this instance we did not try to improve the agreement by refining our description (which could be done, e.g. by introducing strain-rate dependent m). Our aim was rather to demonstrate that a fairly exact prediction of the creep behaviour is possible in a straightforward manner following the above prescriptions. The considered example demonstrates that the one-parameter approach suggested does provide

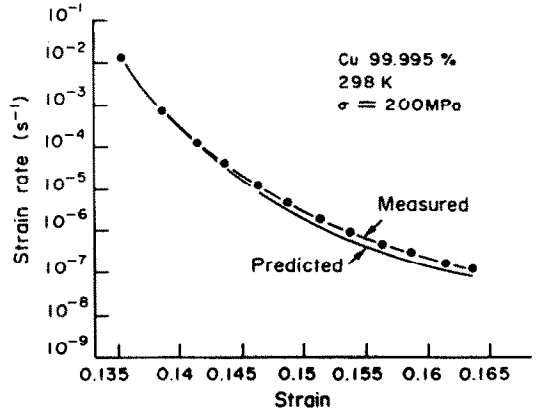


Fig. 8. The measured and the calculated creep curves for Cu.

a good means of converting stress-strain data into creep data.

6. DISCUSSION

The consideration of the two competing models for continuous plastic deformation shows that the one-parameter approach used permits a simple analytical description of the work hardening as well as of the creep behavior in a closed form. Furthermore, the approach is characterized by advantageous universality: if the hardening behavior in the dynamic test is known, the primary (transient) creep behaviour and the steady state creep can be predicted without introducing any adjustable parameters.

Indeed, the quantities θ_0 and k entering the expressions for the creep rate in the two models can be determined from the stress-strain data represented as the θ - σ or $\theta\sigma$ - σ^2 plots as described above. The same plots can be utilized for determining the parameter n . Finally, the parameter m can be found by changing the strain rate. Since the steady state strain rate $\dot{\epsilon}_s$ can be determined with the aid of equation (18c), the entire $\dot{\epsilon}(\epsilon)$ -dependence can be described, for a given initial state ($\epsilon_i, \dot{\epsilon}_i$), without adjustable parameters. A verification of the model giving a more realistic description of plastic deformation for a given material becomes thus possible on the basis of the stress-strain and creep data.

Once verified, the operative model can be used to predict the deformation behavior in both dynamic loading and creep from data obtained for a few (at least two) values of constant strain rate or constant stress. By applying the described procedure at various temperatures and thus determining the temperature dependence of n and m , the variation of plastic behaviour of the material with temperature can be predicted with the aid of the models considered.

Certainly, we are aware of simplifications made in developing the above models. For instance, deviations from the assumed hardening laws observed at small stresses (cf. Figs 1a and 7) (and corresponding to the portions of the stress-strain curves below the

athermal stage, e.g. stage I in single crystals or, even more important, at the yield point in single and polycrystals) have been completely disregarded. One can say that the above description applies to stages II and III of the strain hardening curve where the plots θ vs σ for the KM model and $\theta\sigma$ vs σ^2 for the modified model are linear. Only these linear parts of the curves should be used in processing the data according to the above recipes. For creep, this "truncation" of the assumed hardening laws on the far-from-steady-state side is of no practical importance since due to apparatus limitations only strain rates exceeding $\dot{\epsilon}_i$ by not too many orders of magnitude can be recorded as $\dot{\epsilon}_i$, which implies a fair "proximity" of $\dot{\epsilon}_i$ to steady state. Unfortunately, a large part of the transient creep curve corresponding to the decay of the strain rate down to the *registrable* strain rate $\dot{\epsilon}_i$ is lost for a comparison with theoretical predictions. The strain accumulated during this decay is usually lumped into the so-called "instantaneous strain".

Although we expect that the evolution laws, as given by equations (2a') and (2b') underlying the KM model and the modified model, cover a very broad variety of dislocation processes responsible for macroscopic plastic flow, a "hybrid" model is conceivable of which the two treated above are limit cases. This hybrid model applies to processes in which the dislocation mean free path is determined *both* by dislocation-dislocation interactions *and* by such barriers or sinks as e.g. grain boundaries or sample surface [7]. Assuming the additive operation of the two hardening mechanisms, equations (4) and (8) can be composed to

$$d\rho/d\varepsilon = k + k_1\rho^{1/2} - k_2\rho \quad (32)$$

which, combined with equation (3), results in the following evolution equation for the hardness parameter

$$d\hat{\sigma}/d\varepsilon = (k_2\hat{\sigma}_i^h/2)(\hat{\sigma}_i^h/\hat{\sigma})[1 - (\hat{\sigma}/\hat{\sigma}_i^h)^2 - \kappa(1 - \hat{\sigma}/\hat{\sigma}_i^h)] \quad (33)$$

where

$$\kappa = 2[1 + (1 + 4kk_2/k_1^2)^{1/2}]^{-1}$$

and

$$\begin{aligned} \hat{\sigma}_i^h &= \alpha G b (k_1/k_2)\kappa^{-1} \\ &= \frac{1}{2}\hat{\sigma}_s\{1 + [1 + 4(\hat{\sigma}_s^*/\hat{\sigma}_s)^2]^{1/2}\} \end{aligned} \quad (34)$$

(the superscript *h* standing for "hybrid"). Obviously, for $k = 0$, equation (32) coincides with equation (2a) and for $k_1 = 0$, equation (2b) is recovered.

In the same way as equations (12a,b) for the hardening coefficient in the two considered models were obtained, a corresponding expression in the hybrid model can be found from equation (11). This expression, represented as the stress dependence of the product $\theta\sigma$, reads

$$\theta\sigma = \frac{1}{2}k_2(\sigma_i^h)^2[(1 - \kappa) + \kappa(\sigma/\sigma_i^h) - (\sigma/\sigma_i^h)^2] \quad (35)$$

where

$$\sigma_i^h = \hat{\sigma}_i^h(\dot{\epsilon}/\dot{\epsilon}_i)^{1/m}$$

is the steady state stress for dynamic loading in the hybrid model.

The hardening behavior represented by equation (35) can be compared with that in the KM model and in the modified model which can be viewed as the limit cases of the hybrid model for $k = 0$ (i.e. $\kappa = 1$) and $k_1 = 0$ (i.e. $\kappa = 0$), respectively. While $\theta\sigma$ vanishes at σ tending to the (strain-rate and temperature dependent) steady state stress in all three models, new features are peculiar to the hybrid model at $\sigma \rightarrow 0$. One recognizes that the intercept of the $\theta\sigma$ vs σ plot with the ordinate axis is non-zero and exactly equals that for the modified model viz. $\frac{1}{2}(k\alpha^2b^2)G^2(\dot{\epsilon}/\dot{\epsilon}_i)^{2/m}$. The initial slope of the curve, on the other hand, is identical with that for the KM model, viz. $\theta_0(\dot{\epsilon}/\dot{\epsilon}_i)^{1/m}$. This kind of hardening behavior, illustrated schematically by Fig. 4, was observed in polycrystalline f.c.c. metals [7, 13].

As is obvious from the above considerations, the hybrid model behavior degenerates to the KM model behavior with decreasing k , i.e. with increasing *geometrically determined* mean free path d . Experimentally, this can be investigated by varying the grain size. It is interesting to note that the stress at which a maximum in the dependence of equation (35) is predicted (equal to half the steady state stress as obtained in the KM model) as well as the initial slope of this dependence are insensitive to k (and thus to d) while both the height of the maximum and the steady state stress decrease with d . The corresponding variation laws are trivially derived from equations (34) and (35).

Equation (35) can be integrated without difficulty to yield the stress-strain dependence of the form similar to equations (14a,b)

$$\begin{aligned} \frac{\sigma - \sigma_i^h}{\sigma_i - \sigma_i^h} \left[\frac{\sigma + (1 - \kappa)\sigma_i^h}{\sigma_i + (1 - \kappa)\sigma_i^h} \right]^{1-\kappa} \\ = \exp[-(\varepsilon - \varepsilon_i)/\varepsilon_r^h] \end{aligned} \quad (36)$$

where the relaxation strain is given by

$$\varepsilon_r^h = 2k_2^{-1}(2 - \kappa)^{-1}.$$

Equation (14a) for the KM model and equation (14b) for the modified model follow from equation (36) at $\kappa = 1$ (i.e. $k = 0$) and $\kappa = 0$ (i.e. $k_1 = 0$), respectively.

Also, by combining equation (11) taken at $\sigma = \text{const.}$, with equation (33) a differential equation giving the strain dependence of the strain rate in creep can be produced. This equation is integrable as well. The resulting relationship between the strain rate and strain in constant stress creep reads

$$\begin{aligned} \frac{[1 + (kab/k_1)(G/\sigma)](\hat{\sigma}_0/\sigma)(\dot{\epsilon}/\dot{\epsilon}_0)^{1/n+1/m} - 1}{[1 + (kab/k_1)(G/\sigma)](\hat{\sigma}_0/\sigma)(\dot{\epsilon}/\dot{\epsilon}_0)^{1/n+1/m} - 1} \\ = \exp[-(\varepsilon - \varepsilon_i)/(\varepsilon_r^{\text{creep}})^h] \end{aligned} \quad (37)$$

where the relaxation strain $(\epsilon_r^{\text{creep}})^h$ is given by

$$(\epsilon_r^{\text{creep}})^h = -\frac{n}{m+n} (\sigma/\theta_0) [1 + (kab/k_1)(G/\sigma)]^{-1}.$$

It is easily recognized that equation (19a) (KM model) and equation (19b) (modified model) follow from equation (37) at $k = 0$ and $k_1 = 0$, respectively.

The results of the present investigation show that simple qualitative features of the hardening behavior of metals and alloys (and, more generally, of crystalline materials) can be used to decide whether one of the two one-parameter models discussed, or their hybrid form, is applicable for a phenomenological description of stress-strain and creep data. A quantitative analysis can be then carried out with the model chosen. On the basis of this analysis, the creep behavior can be predicted from the stress-strain behavior. This program has been successfully applied for converting the data on the hardening behavior of polycrystalline Cu under dynamic loading (which exemplifies a clear-cut realization of the Kocks-Mecking model) into creep data.

The two models considered seem to embrace a great variety of experimentally encountered situations so that the above approach appears rather "universal". That is why we do not refer to any particular class of materials or to specified experimental conditions (although we have in mind as a practically important illustration the low-temperature deformation of f.c.c. metals, cf. Section 5). The approach developed is a mechanistic one. The consideration of the underlying dislocation mechanisms provides it, nevertheless, a large degree of plausibility and physical meaningfulness. It is for future experiments to decide how "universally" this one-parameter approach can be applied.

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APPENDIX

1. Estimating the error associated with using the UTS

The criterion for the onset of necking [4, 14] reads

$$(\partial \ln \sigma / \partial \epsilon)_\epsilon = 1. \quad (\text{A1})$$

By combining equations (12a) and (A1) we get for the KM model

$$\sigma_{\text{UTS}} = \sigma_s / [1 + (\sigma_s / \theta_0)(\dot{\epsilon} / \dot{\epsilon})^{-1/m}] = \sigma_s / (1 + 2k_2^{-1}). \quad (\text{A2})$$

Substituting σ_{UTS} instead of σ_s in equation (22), i.e. utilizing

$$1/n_{\text{UTS}} + 1/m = (\partial \ln \sigma_{\text{UTS}} / \partial \ln \dot{\epsilon})_T,$$

gives the value of n , denoted as n_{UTS} , which reads

$$n_{\text{UTS}} = n(1 + 2k_2^{-1}). \quad (\text{A3})$$

Consequently, $\Delta n = n_{\text{UTS}} - n = 2k_2^{-1}n$, in accordance with equation (25). For the modified model, equation (A1) is combined with equation (12b) to give as the UTS

$$\begin{aligned} \sigma_{\text{UTS}}^* &= \sigma_s^* / [1 + 2(k\alpha^2 b^2)^{-1} \\ &\quad \times (\sigma_s^* / G)^2 (\dot{\epsilon} / \dot{\epsilon})^{-2/m}]^{1/2} \\ &= \sigma_s^* / (1 + 2k_2^{-1})^{1/2}. \end{aligned} \quad (\text{A4})$$

The substitution of σ_{UTS}^* in place of the steady state stress in equation (22), i.e. utilizing

$$1/n_{\text{UTS}}^* + 1/m = (\partial \ln \sigma_{\text{UTS}}^* / \partial \ln \dot{\epsilon})_T,$$

gives an apparent value of n , denoted here as n_{UTS}^*

$$n_{\text{UTS}}^* = n^*(1 + 2k_2^{-1}). \quad (\text{A5})$$

The relative error, $\Delta n^* / n^* = (n_{\text{UTS}}^* - n^*) / n^*$ is given again by equation (25).

2. Estimating the error associated with using the minimum creep rate

The tensile instability criterion for creep at constant stress [14] which can be represented in our notation as†

$$(\partial \ln \dot{\epsilon} / \partial \epsilon)_{\sigma, T} = -n \quad (\text{KM model}) \quad (\text{A6a})$$

$$(\partial \ln \dot{\epsilon} / \partial \epsilon)_{\sigma, T} = -n^* \quad (\text{modified model}) \quad (\text{A6b})$$

defines the minimum creep rate $\dot{\epsilon}_{\text{min}}$ or $\dot{\epsilon}_{\text{min}}^*$ in the KM model or the modified model, respectively. By combining equation (A6a) with (16a) and equation (A6b) with (16b) the following expressions are obtained for the two models

$$\dot{\epsilon}_{\text{min}} \cong \dot{\epsilon}_s \left[1 - \frac{n}{m} (\sigma / \theta_0) \right]^{-n} \quad (\text{KM model}) \quad (\text{A7a})$$

$$\begin{aligned} \dot{\epsilon}_{\text{min}}^* &\cong \dot{\epsilon}_s^* \left[1 - 2 \frac{n^*}{m} (k\alpha^2 b^2)^{-1} (\sigma / G)^2 \right]^{-n^*/2} \\ &\quad (\text{modified model}). \end{aligned} \quad (\text{A7b})$$

Now, using the minimum creep rate instead of the steady state strain rate in equation (26), i.e. defining the powers n_{min}

†It should be noted that the relation $(\partial \ln \dot{\epsilon} / \partial \ln \sigma)_\epsilon = n$, rather than equation (1'), underlies the analysis of Ref. [14] and thus equation (A6a). However, its use appears reasonable in view of the proximity of the instability to steady state. A similar argument applies to equation (A6b).

and n_{min}^* from

$1/n_{min} + 1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon}_{min})_T$ (KM model) (A8a)

$1/n_{min}^* + 1/m = (\partial \ln \sigma / \partial \ln \dot{\epsilon}_{min}^*)_T$ (modified model) (A8b)

we get

$\Delta n = n_{min} - n \cong \frac{n^2}{m} (\sigma / \theta_0)$ (A9a)

$\Delta n^* = n_{min}^* - n^* \cong 2 \frac{(n^*)^2}{m} (k\alpha^2 b^2)^{-1} (\sigma / G)^2$. (A9b)

The evident smallness of the terms

$\frac{n}{m} (\sigma / \theta_0)$

and

$2 \frac{n^*}{m} (k\alpha^2 b^2)^{-1} (\sigma / G)^2 = 2\alpha^{-2} (n^* / m) (d / b) (\sigma / G)^2$

as compared to unity has been used here.

Equations (A9a) and (A9b) are rewritten in the text as equations (29a) and (29b), respectively.